
Classical-CV chart digitizer --- ordered pipeline (OpenCV + numpy + PaddleOCR ONNX)
Screenshot (RGBA PNG)
1. preprocess
EXIF-transpose, RGB->BGR,
resize into [1200,2400] work band
2. color_segmentation
HSV inRange: blue (BBT),
orange (Ratio), purple (Level)
+ morphological close
3. plot_region
ink bbox U gridline band
-> PlotRegion(x0,y0,x1,y1)
4. axis_columns (PaddleOCR)
margin word-boxes -> column clusters
OCR -> RANSAC line fit per column
classify ratio/level/bbt_f/bbt_c
5. day_axis
autocorrelation pitch + date row
+ gridlines -> DayGrid(cells)
N_DAYS=35 left-packed
6. marker_detection
white-blob + colored-ring;
resolve LH color (orange vs purple);
discrete OR dense line-sampling
Deterministic: no learned weights.
Every threshold is an explicit,
auditable constant.
7. table_extract
calendar/CD/DPO/Sex/CM/
Symptoms/hCG rows + cells
8. guardrails
snap axis labels to AP grid;
linear gap interpolation
(present mask untouched)
9. quality
weighted per-stage confidence;
status: extracted / low_conf /
not_a_chart (axis is a gate)
ChartResult
value[2,35] in [0,1], present[2,35],
scale_idx (C/F), table, confidence,
status, interpolated, debug overlay